

Mid Semester Examination

Examination: V Sem. B.Tech (MNE)

Semester: Monsoon Semester

Subject: Mine Planning and Economics (MNC301)

Note-All questions are compulsory

Session: 2023-24

Time: 2 Hours

Max. Marks: 30

Q. No.	Questions	Marks
1	From the given sequences find out which is more profitable sequence of excavation Year 0 Year 0 Year 1 $V_1 = -\\$5$ Year 1 $V_1 = -\\$5$ Year 2 $V_2 = -\\$5$ Year 2 $V_2 = \\$4$ Year 3 $V_3 = \\$10$ Year 3 $V_3 = \\$4$ Year 4 $V_4 = \\$10$ Year 4 $V_4 = \\$7$ Rate of interest is 10%	4
2	Find out the milling rate, mine production rate, mine life and overall stripping ratio from the information given below; Metal demand=6000 te/years; ore reserve=8×10^6 te; Tonnes to be mined (overburden and ore)= 32×10^6; Avg. grade=0.9%; mill recovery=80%; refinery recovery=90%; working days=250 per year	4
3	Calculate overall bench slope angle from the following information; Bench face angle=75°; height of bench= 30m; width of safety bench= 25m; width of working bench= 50m, no. of safety benches=3; no. of working benches=3	3
4	A mine is having 200million tones of limestone reserve and is planned to produce 20,000 tonnes/ day from multiple benches of a mine with bench height of 10m and drill hole diameter of 150mm, As a planning engineer you are asked to give the requirements of drilling machine, explosives, shovels ($5m^3$) and the life of the mine. Assume further required information.	2+2+2 +1
5	List drills available for exploration purpose and explain the resource estimation techniques	1+3
6	Write short notes on the followings - Floating cone method for ultimate pit configuration - Factors affecting the choice of equipment in a mine - Haul road geometry - Components of mine planning	4*2